

CAAM/STAT 31410 Applied Dynamical Systems Autumn 2025 Syllabus

Class Meeting: TTh 11 AM - 12:20 PM, Kersten 101

Instructors- Nisha Chandramoorthy (nishac@uchicago.edu)
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Course Assistant – Matthew Oline (moline@uchicago.edu)

Course Webpage – Canvas : <https://courses.uchicago.edu/> (Login with CNet ID)

- Check often! Handouts/notes/HWs/announcements are all posted there.

Course Description: This course is an introduction to dynamical systems in the setting of nonlinear ordinary differential equations. The focus is on basic existence-uniqueness theorems for ODEs, stability of equilibria and periodic solutions, existence and uniqueness of invariant manifolds, methods of bifurcation theory including center manifold reduction, normal form transformation and global bifurcations, as well as chaotic dynamics. While geometric perspectives will be emphasized, assignments will also introduce approximation methods that will guide explorations via numerical simulation.

Course Prerequisites: This course assumes students have a background in ordinary differential equations and linear algebra at the undergraduate level, and an interest in mathematical modeling for applications. The course requires some minimal experience with, or willingness to learn, simple programming for solving ordinary differential equations, e.g. using built-in ODE solvers in Python, Matlab or Mathematica.

Recommended Reference Textbook: *Differential Dynamical Systems*, by James D. Meiss.

Office Hours:

- Nisha Chandramoorthy: Tues. & Thurs.: 12:30-1pm in Jones 122B, and by appointment
- Mary Silber: Mon. 4-5pm in Jones 116A, and by appointment
- Matthew Oline: Mon. 10-11am in Pick 319, and Wed. 4:30-5:30pm in Harper Memorial 151

Grade Components

- Homework (40%): Expect 5 substantial problem sets.
- Midterm Quiz (20%): November 18, in class.
- Final Exam (40%): December 11, 10am-12pm, in class.

Homework Policies

- Please use the **online submission** of homework feature on *Canvas*:

<https://courses.uchicago.edu/> (log in with CNet ID and password)

- You can **type/LaTeX** your homework and submit a pdf, or
 - **scan** your **handwritten** homework and then upload it to *Canvas*.
- For each assignment, please combine your work (text and graphs) in a **single file** for submission, not multiple ones.
- You can submit a single homework multiple times, before the due time. So if you want to correct a mistake in your submitted homework, you can just correct it and submit it again. We will grade the last version that is submitted prior to the submission deadline.
- **Late homework** will not necessarily get graded, or, even if accepted and graded, it may be marked down for being late.
- **Homework Collaboration:** You are encouraged to discuss course material and homework with other students, with the following restrictions:
 - You must **do the final write-up independently in your own words**, and **do your own computer work (codes, plots)**.
 - If you receive substantial help on a problem, you must acknowledge it. This includes any online resources you use.
- Problem solutions must be coherent and legible. Graphs must be properly labeled.
- Please show your work with enough detail that someone else in the class could reproduce your solution by simply following along.